S P E C I F I C A T I O N No. 72512461

Wülfrath, 13.03.2025
Person in charge: Sven Borghoff
Inter no.: 107812
Inquiry-/Project no.: HFE
Order no.:
Purchase order no.:
REVISION 01

Client code: 15729
Chemac Inc.

Mrs. Kimberley Stuart
USA-77511 Alvin / TX

Your application data	: Metering		
Liquid	: Novec 7000, Novec 7200 Please check additionally the chemical resistance of the materials!		
Solids	: none		
Temperature	: -100 to +30	°C,	ambient temperature 15 - 35 °C
Density	: 1,39 - 1,75	kg/l	
Viscosity	: 0,39 - 39,0	mPa s	
Inlet pressure	: 1,00	bar abs	Inflow
Outlet pressure	: 3,00-11,00	bar abs	Max. permissible system pressure: 16 bar abs.
Differential pressure	: 2,00-10,00	bar	maximum
Capacity	: 0,500-4,000	l/min, (30,0-240,0 l/h)	
r.p.m.	: c. 200-2150	1/min, for mentioned viscosity range	
Operating time	: 24	h/day	

It is possible to clean the pump in place (CIP, please see data sheet)

Your gear pump head, magnetic drive - Series 2

Gear width		: 8 mm - 12 teeth
Materials:	Body	: Stainless steel 1.4404
	Shafts	: Stainless steel 1.4571 (with DLC coating)
	Driving/driven gear	: Ni-alloy / Ni-alloy
	Bearings	: Slide bearings - Carbon (Sb-impregnated)
	Seals	: PTFE (FDA conform, USP Class VI)
	Magnet type - G-magnet	: shrouded in stainless steel 1.4571
Ports		: welded pipe socket 12 mm male resp. 10mm female diameter, with screw cap VCR 1/2", SW 1 1/16", welded laterally
Model:	2M-J/8/G-SS/VCR/DLC/	

Quantity: 1
Item ID: 2M-J/8/G-SS/VCR/DLC/
Item no.:

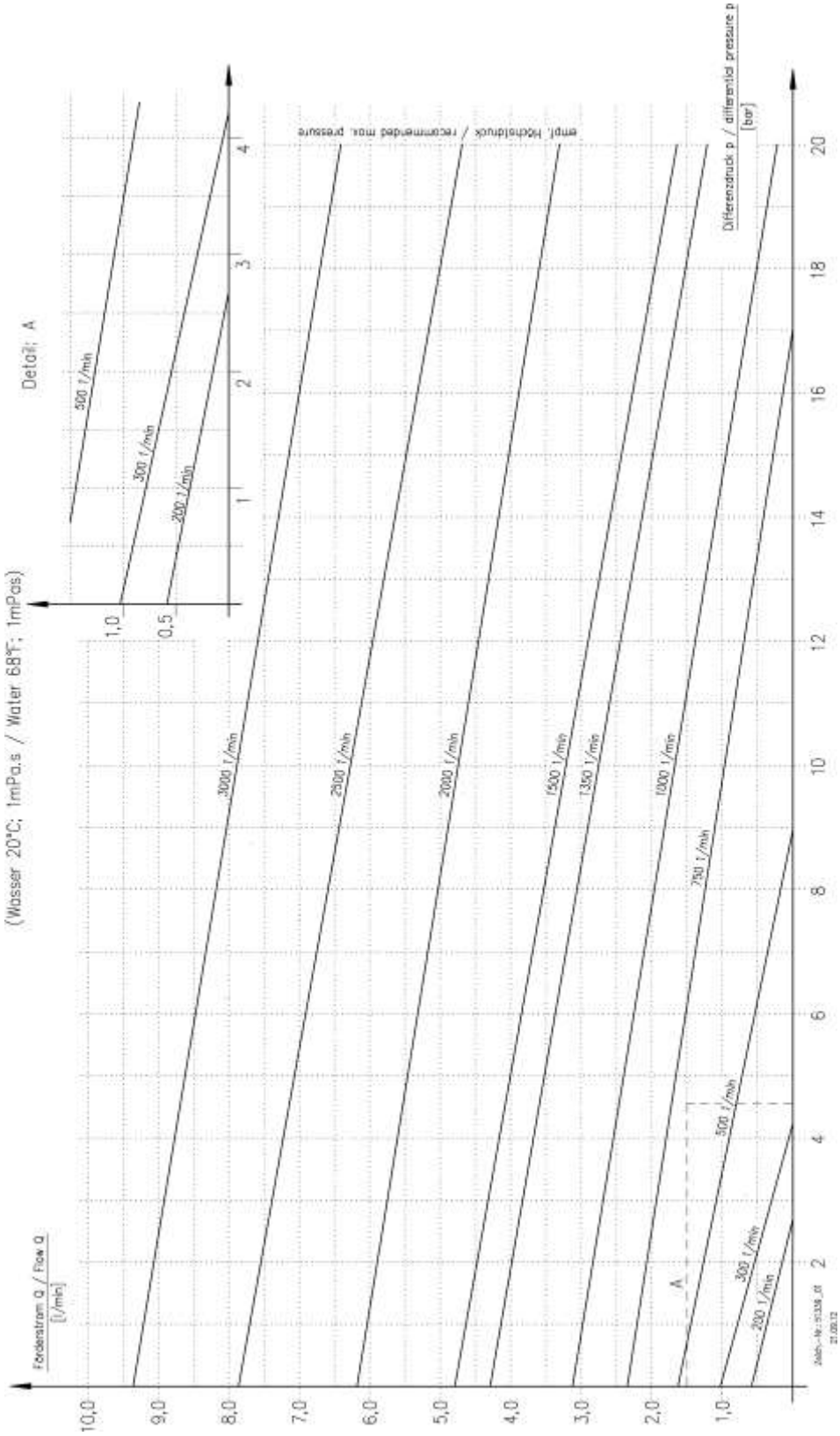
Pump head/s dried

Accessories:
Motor connecting parts for pumps of series 2 (low temperature design), for motor 143TC/145TC, (each) consisting of:
- bell housing made of aluminium (mod.)
- adapter flange made of aluminium (mod.)
- driving magnet type G, boring 22,22mm
- heat insolating adapter (GFK)
- mounting screws

Quantity: 1
Item ID: G/GFK/NEMA 143TC/145TC-Flansch-ALU-Tieftemperatur
Item no.: 31297

Electric Motor: Baldor Severe Duty, 1HP, 3PH/230-460V/60Hz, Inverter ready, TEFC, NEMA 143/45TC, C-Face, foot mounted, Speed range 50-2700rpm, C1D2, VFD Ready, Inlet on Bottom

Kennlinie:	Baureihe 2
Zahnbreite / Zähnezahl:	8mm / 12 Zähne
Performance curve:	Series 2
Gear width / number of teeth:	8mm / 12 teeth



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Pumpentechnik
Dosiertechnik
Kupplungstechnik

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Cleaning the GATHER Magnetic Drive Pump

A pump that is no longer functioning cannot be cleaned when mounted in the system!

Note: Please follow the operating manual, when shut down and remove the pump.

In such a case shut the pump down and remove the pump head. Disassemble the pump head to the extent possible – the magnet cap must be removed in any case – and clean the head, for example manually or in an ultrasonic bath.

CIP – Cleaning in Place

The special design of the GATHER pump allows CIP pump cleaning (i.e. cleaning in place): During operation the bearings and fluid conveying elements (gearwheels or impellers) of the pump are lubricated and cooled via internal purge ducts by the fluid conveyed. Because the magnet space is purged as well a continuous exchange of fluid is achieved throughout the pump when aqueous liquids are pumped.

This is absolutely necessary for CIP cleaning purposes and enables the pump to be cleaned (using e.g. hot water, solvents or other liquids) without having to dismount it.

For pump cleaning the respective cleaning agent must be fed to the pump. The pump should be operated at a minimum motor speed of (n_{min} = 2,000 min^{-1} for gear pumps, n_{min} = 4,000 min^{-1} for centrifugal pumps of series 1, and n_{min} = 2,750 min^{-1} for centrifugal pumps of series 2. To make sure the pump can be purged or flushed through inside, a differential pressure of Δp_{min}= 1.0 bar has to be built up between delivery and intake side. Pump cleaning roughly takes 5 to 10 minutes (depending on pumping fluid).

Prior to first cleaning it must be verified that the materials are resistant to the cleaning agent used. The customer or Plant Owner is responsible for the selection and application of the solvent.

The CIP cleaning temperature must not exceed the maximum operating temperature stated in the pump-related specification!

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Cleaning of the pump from higher viscose liquids

To be able to clean the pump of fluids having a viscosity in excess of 100 mPa s at the cleaning temperature, the pump requires an additional purge connection (type: „/SP/“, „/SP2/“ or „/SIP/“).

Cleaning of the pump

1. Rinse the pump with cleaning liquid approx. for 10 minutes (see CIP cleaning). Meanwhile the bypass valve must be closed.
2. Subsequently, close the outlet pressure side of the pump with a shutoff device and open the bypass valve. Rinse the pump again approx. for 2-5 minutes.
3. Afterwards close the bypass valve again and open the outlet pressure side of the pump. Rinse the pump again (see 1.) approx. for 5-10 minutes.

Sterilization

Irrespective of the sterilization method to be adopted, sterilization itself must always be included in the pump-related specification. Normally, pump heads with threaded connections cannot be sterilized.

Basically, it must be ensured the pump parts in contact with the fluid reach a temperature higher than 121°C during sterilization.

Sterilization in a heating cabinet:

If the pump is to be sterilized in a heating cabinet, shut the pump down and remove the pump head. Do not disassemble the pump head; take care not to close off the connections and place it in a heating cabinet for a sufficiently long time. Observe all safety notes pertinent to the heating cabinet and handling of hot components.

Sterilization using hot water:

If the pump is to be sterilized in mounted condition using superheated, pressurized water (at least 125°C inside the pump), the pump can be purged as per CIP cleaning and sterilized in this manner.

Sterilization with Steam:

If the pump is to be sterilized in mounted condition using steam (135°C) (SIP- sterilization in place), an additional steam bypass system is needed for the pump (type: „/SIP/“).

Further information on this design and inline sterilization can be obtained from our Head Office.

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